Naive substring search and KMP

First-occurrence search over bytes

Ethan Goldman

Advanced Algorithms 41052

The search contract

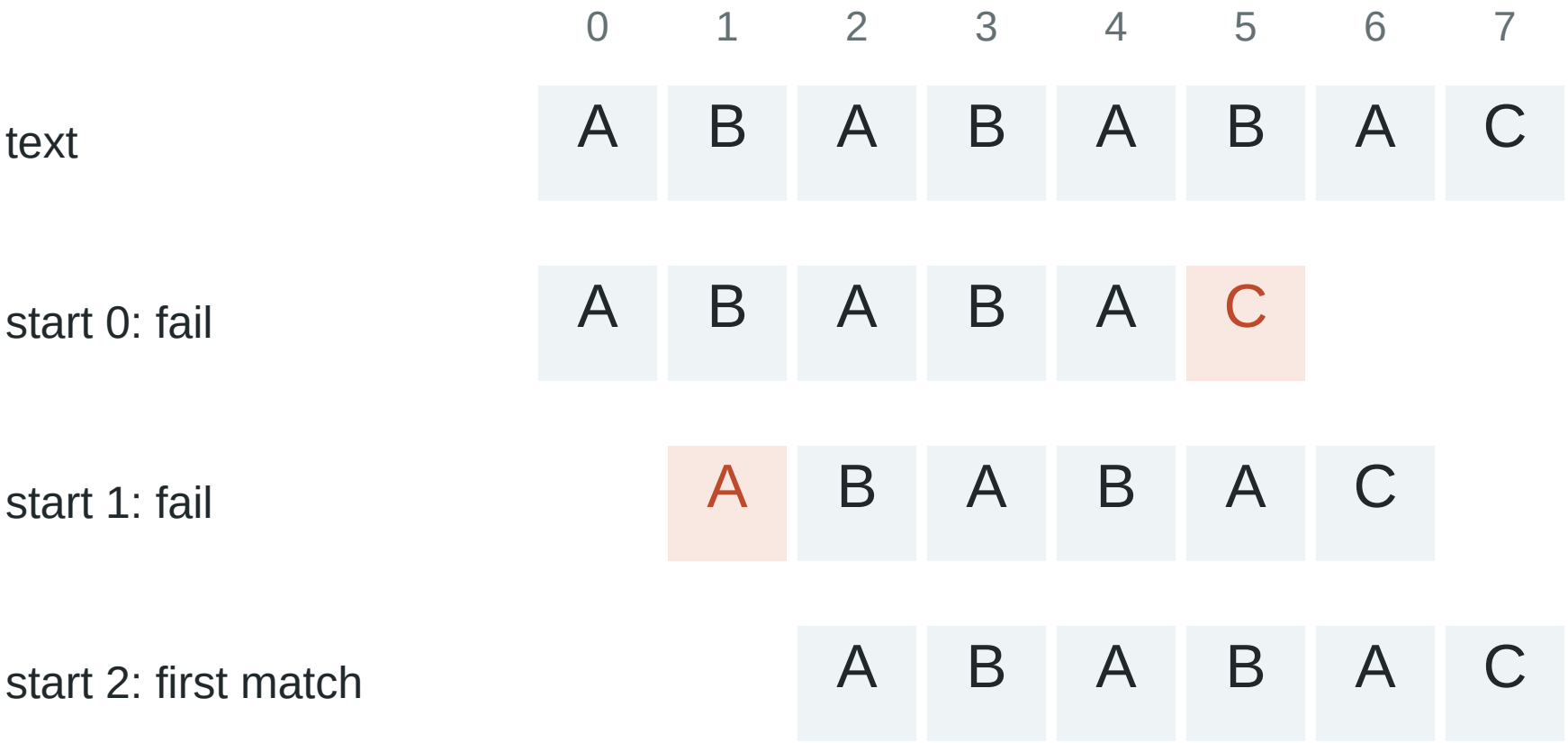
Return the first matching byte index

Empty pattern: 0 Absent pattern: npos

Zero bytes are valid. Inputs stay unchanged.

ABABABAC with ABABAC returns 2

Naive search: every valid start



13 search equality tests. Worst case: (n - m + 1)m.

KMP: the LPS table

LPS = longest proper prefix that is also a suffix

index	0	1	2	3	4	5
pattern	A	B	A	B	A	C
LPS	0	0	1	2	3	0

ABABA keeps ABA as both prefix and suffix

At C, try border lengths 3, 1, then 0

O(m) preprocessing time and O(m) auxiliary space

KMP fallback preserves a valid match

Keep i = 5. Compare B with pattern[3].

Valid retained prefix: ABA. First match: 8 - 6 = 2.

Predictions saved before timing

Random early mismatches may favour naive

Long partial matches should favour KMP

Beginning matches expose preprocessing cost

Algorithm	Worst-case time	Auxiliary space
naive	O((n - m + 1)m)	O(1)
KMP	O(n + m)	O(m)

A controlled measurement

363 instances, nine timed batches per instance

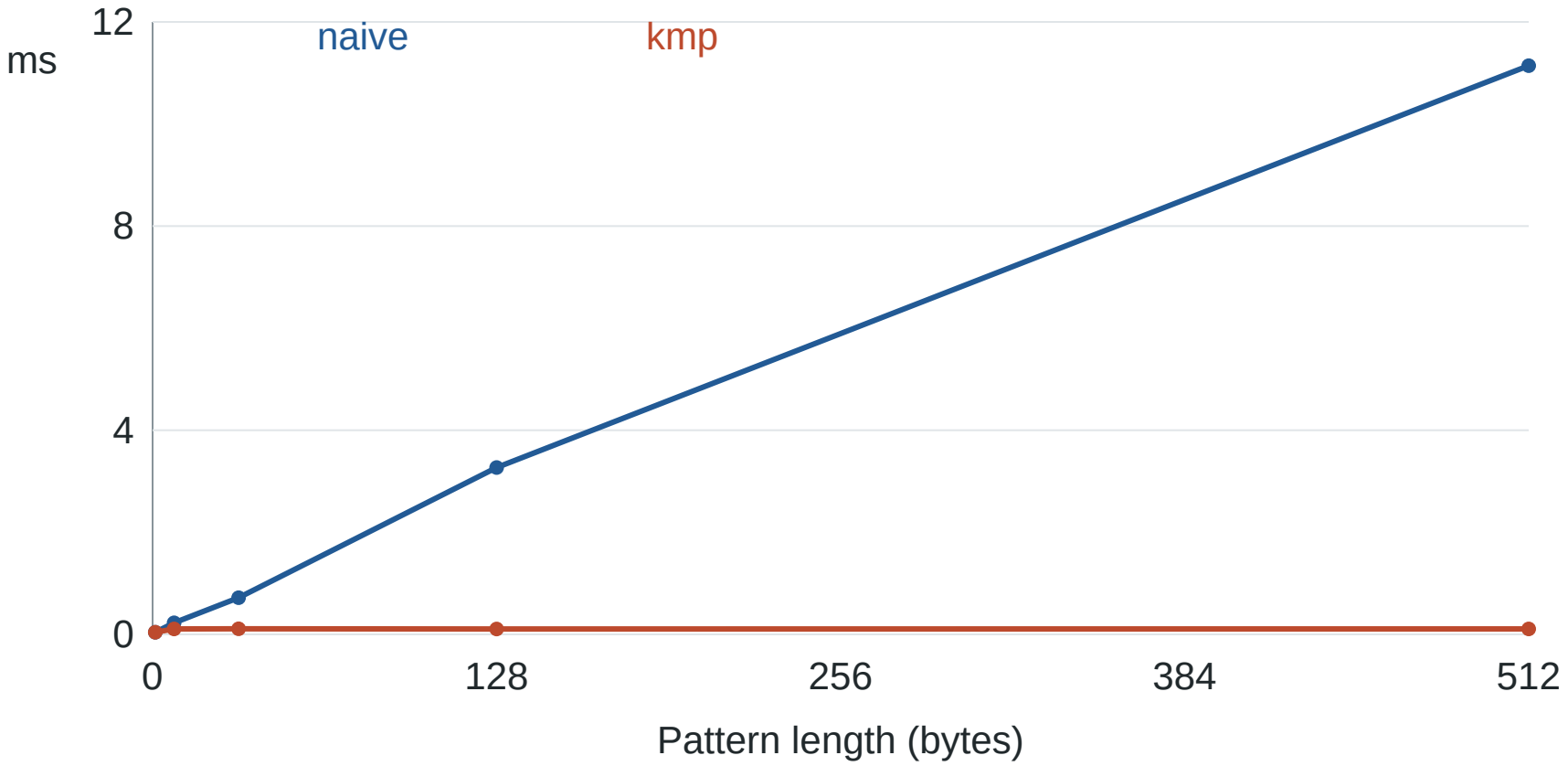
Same inputs, verified first positions, alternating order

KMP table construction included on every call

M1 Pro, 16 GiB, macOS, Apple Clang 21, C++17 -O2

Long partial matches reward KMP

Repeated text, absent pattern, n = 65,536



104.48x

faster at
m = 512

Dots: median of three instance medians. Nine batches per instance. IQR in report.

Comparison counts test the explanation

Naive: 33,292,800 equality tests

KMP: 131,582, including preprocessing

Count ratio 253.02; runtime ratio 104.48

Algorithm	Time (microseconds)	Across-instance IQR
naive	11143.000	11135.450-11203.200
KMP	106.656	106.227-107.307

Almost equal counts, different runtimes

Text: a repeated 65,536 times

Pattern: ab followed by 510 a bytes

Both absent searches, m = 512

Algorithm	Equality tests	Time (us)	IQR (us)
naive	130,050	61.691	61.319-61.781
KMP	132,091	107.511	107.373-107.688

KMP is 1.74 times slower. Equalities omit surrounding work.

Limits of the evidence

One measured machine and compiler

Three instances per configuration

Prose windows come from one repeated excerpt

The choice depends on the workload

KMP protects against repeated partial matches

Naive remains competitive when failures are cheap

Next: search many texts with one reusable pattern

github.com/egoldman2/substring-search